Customer Shopping Behaviour Analysis

1. Project overview:

This project looks at how customers shop by studying the information from 3,900 purchases of products. The project on customer shopping behavior wants to find out how customers spend money, group customers see what products customers like and check if customers like to subscribe to things. The customer shopping behavior project does this to give businesses the information they need to make decisions, about the customer shopping behavior project.

2. Data Summary:

- Rows: 3,900
- Columns: 18
- Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. EDA(Exploratory Data Analysis)

Begin with Data preprocessing with Python. Following steps were carried out:

❖ **Data Loading:**

Data was imported using pandas.

❖ **Data Exploration:**

Used df.info ( ) to check structure and .describe( ) for summary statistics.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN

- **Missing Data Handling**

  Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.

- **Column standardization:**
  Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
  - ✓ Created **age_group** column by binning customer ages.
  - ✓ Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:**

  Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.

- **DataBase Integration:**

  Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis Using SQL

Different structured analysis in PostgreSQL were performed to answer some interesting business question:

Q1. What is the total revenue generated by male vs female customers?

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Q2. Which customers used a discount but still spent more than the average purchase amount?

	customer_id bigint	purchase_amt bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68

Q3. Which are the top 5 products with the highest average review rating?

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Q4. Compare the average Purchase amount between standard and express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Q5. DO subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Q6. Which 5 products have the highest percentage of purchases with discounts applied?

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

Q7. Segment customers into new, Returning and loyal based on their total number of previous purchases and show the count of each segment

	customer_segment text	Number of Customer bigint
1	Loyal	3116
2	New	83
3	Returning	701

Q8. What are the top three most purchased products within each category?

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169

Q9. Are customers who are repeat buyers(more than 5 previous purchases) also likely to subscribe?

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Q10. What is the revenue contribution of each age group?

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-Aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard using Power BI

Finally a dashboard was built in Power BI to present insights visually.

6. Business Recommendations:

- Target high spending discount users- personalized coupons instead of valuable customers.
- Focus marketing budget on the **higher-spending gender segment**
- Target repeat buyers with **subscription offers**
- Focus marketing efforts on **highest revenue-generating age group**